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Exp No.: 7 Endorsement Policy Configuration in Hyperledger Fabric

Aim

To configure and implement endorsement policies in Hyperledger Fabric and verify how multiple organizations endorse transactions before they are committed to the blockchain ledger.

Software Requirements

- Kali Linux / Ubuntu
- Docker
- Docker Compose
- Node.js (Version 18 or later)
- npm
- Git
- Hyperledger Fabric Samples
- Hyperledger Fabric Binaries
- Hyperledger Fabric Docker Images
- Visual Studio Code (Optional)

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB Free Disk Space
- Internet Connection

Theory

An Endorsement Policy defines which organizations must approve (endorse) a transaction before it can be committed to the ledger. Hyperledger Fabric uses endorsement policies to ensure that transactions are validated according to predefined business rules.

For example, `OR('Org1MSP.peer','Org2MSP.peer')` means that either Org1 or Org2 can endorse the transaction. Similarly, `AND('Org1MSP.peer','Org2MSP.peer')` means that both organizations must endorse the transaction before it becomes valid. Endorsement policies improve security, trust, and transaction integrity in permissioned blockchain networks.

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Procedure

Step 1: Navigate to the Test Network

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Fabric Network

```
./network.sh up createChannel -ca
```

Step 3: Deploy Chaincode with OR Endorsement Policy

Deploy the Asset Transfer chaincode with a policy allowing either organization to endorse.

```
./network.sh deployCC -ccl javascript -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-javascript \
  -ccep "OR('Org1MSP.peer','Org2MSP.peer')"
```

Step 4: Verify Chaincode Commitment and Approvals

Check that both organizations have approved and the chaincode is committed.

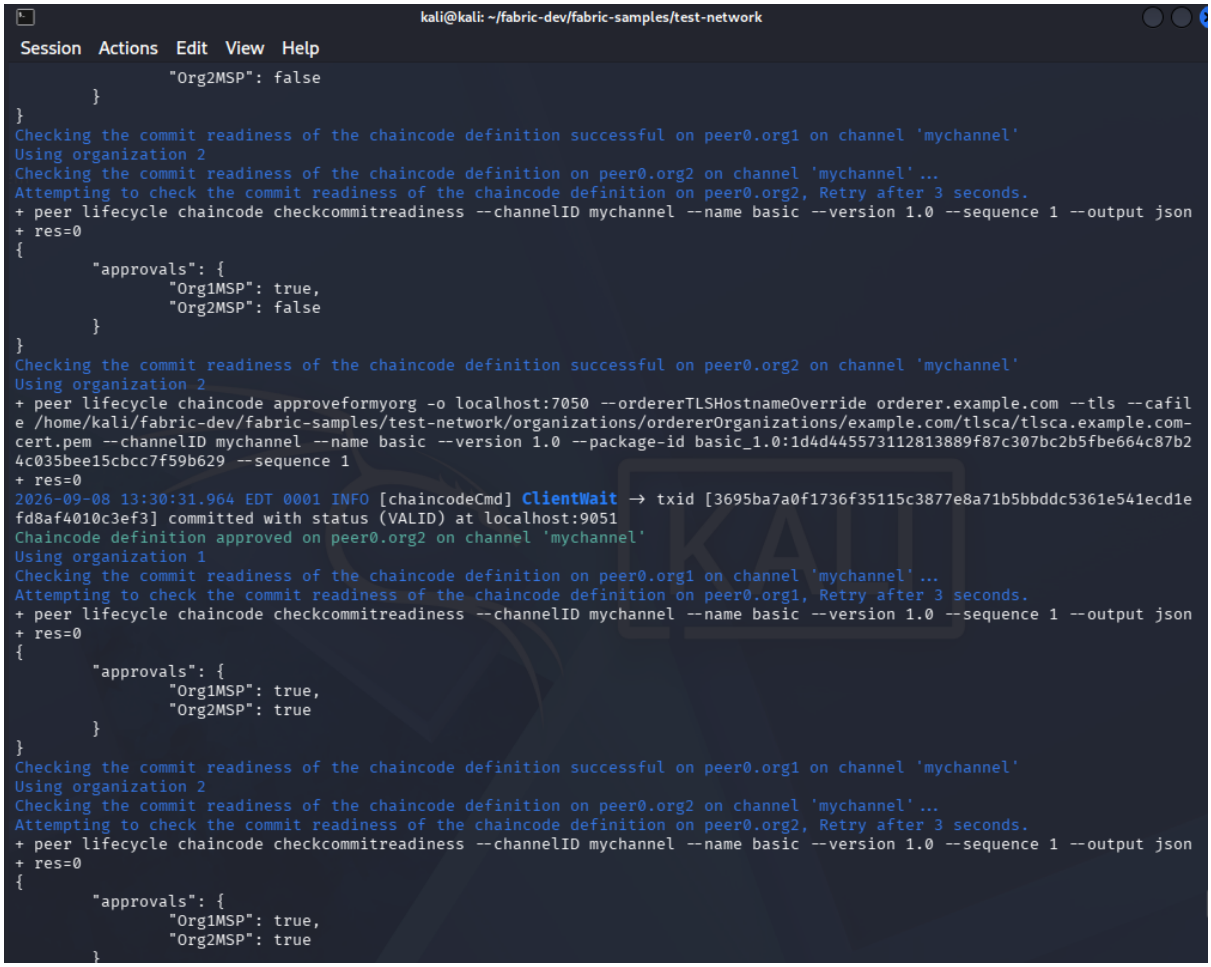

Figure 1: Chaincode approval, commit readiness check, and commitment on mychannel

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Step 5: Initialize the Ledger

```
peer chaincode invoke ... -c '{"function":"InitLedger","Args":[]}'
```

Step 6: Query All Assets

```
peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
```

Step 7: Stop the Network

```
./network.sh down
```

Step 8: Restart and Deploy with AND Endorsement Policy

Restart the network and deploy the chaincode requiring both organizations to endorse.

```
./network.sh up createChannel -ca
./network.sh deployCC -ccl javascript -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-javascript \
  -ccep "AND('Org1MSP.peer','Org2MSP.peer')"
```

Step 9: Submit a Transaction (CreateAsset)

Create asset25 — transaction requires endorsement from both Org1 and Org2.

```
peer chaincode invoke ... \
  -c
'{"function":"CreateAsset","Args":["asset25","Blue","10","Arun","1200"]}'
```

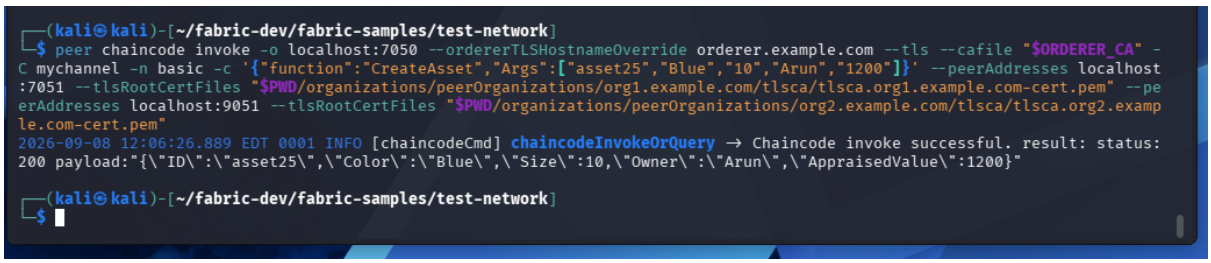

Figure 2: CreateAsset transaction endorsed successfully by both organizations

Step 10: Verify the Asset

```
peer chaincode query -C mychannel -n basic -c
'{"Args":["ReadAsset","asset25"]}'
```

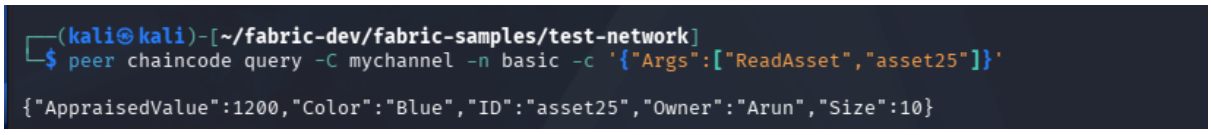

Figure 3: ReadAsset confirms asset25 was created with the correct attributes

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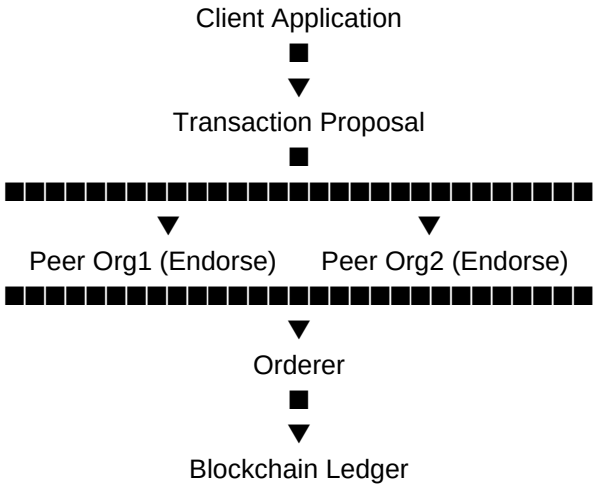
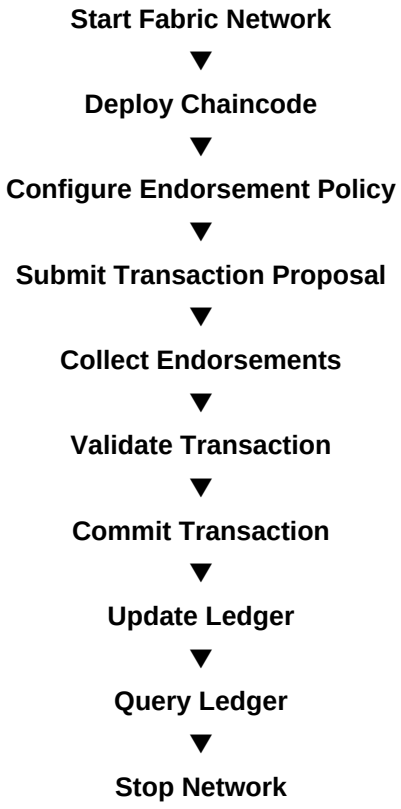
Step 11: Stop the Network

```
./network.sh down
```

Architecture Diagram



Flow Diagram



Result

The endorsement policies were successfully configured in the Hyperledger Fabric network. Transactions were validated according to the specified endorsement rules, demonstrating how Hyperledger Fabric ensures trust, integrity, and secure transaction processing through endorsement policies.